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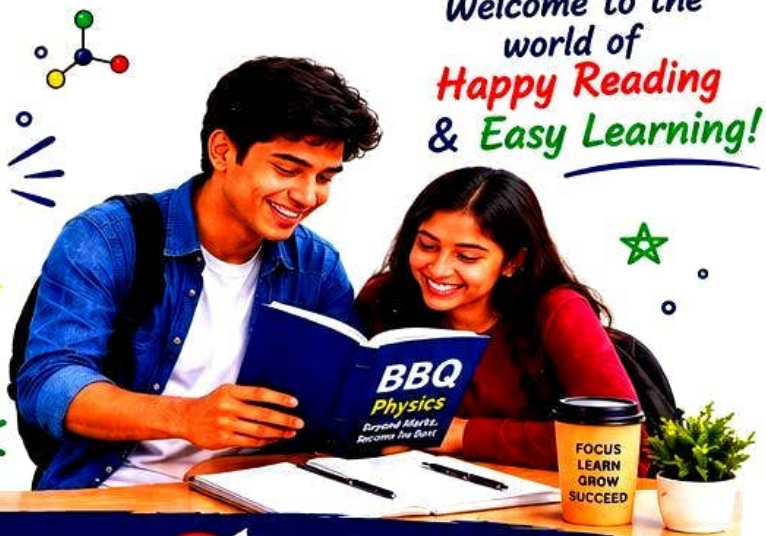
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1. This ' Baby Bullet Book' is prepared keeping in view of the needs of Rising Stars (Pass Aspirants).
2. This is a 'Minimum Learning Reivision Material' useful for Rising Stars.
3. This Booklet contains mostly important expected questions.
4. A few answers are 'reduced or simplified' to make them ' easy to read ' and 'easy to wirte '.
5. The Answers are written as per the ' general guidelines of the Scheme of Valuation'.
6. But one 'should not expect full marks' for each and every Bullet Answer.
7. Various features such as Hit List, Selection of Q's & orderly arrangement of sections, questions. Model Guess Papers etc are prepared just keeping in view of the needs of Rising Stars.
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# IPE BLUE PRINT OF SR PHYSICS

| CHAPTER (Total Marks) | OMQ(1M) 9×1= 9 | VSAQ(2M) 14×2 = 28 | SAQ(4M) 8×4 = 32 | LAQ(8M) 2×8 = 16 | Total 85 M |
|---|---|---|---|---|---|
| 1. Electric Charges and Fields (7 M) | 1 | 2 | 4 | - | 7 |
| 2. Electrostatic Potential (10 M) | - | 2 | 4+4 | - | 10 |
| 3. Current Electricity (7 M) | 1 | 2 | 4 | | 7 |
| 4*. Moving Charges & Magnetism (10 M) | - | 2 | - | 8 | 10 |
| 5. Magnetism and Matter (7 M) | 1 | 2 | 4 | - | 7 |
| 6. Electromagnetic Induction (7 M) | 1 | 2 | 4 | - | 7 |
| 7. Alternating Current (7 M) | 1 | 2 | 4 | - | 7 |
| 8. Electromagnetic Waves (3 M) | 1 | 2 | - | - | 3 |
| 9*. Ray Optics & Optical Instruments (10 M) | - | 2 | - | 8 | 10 |
| 10. Wave Optics (7 M) | 1 | 2 | 4 | - | 7 |
| 11. Dual Nature of Radiation & Matter (10 M) | - | 2 | 4+4 | - | 10 |
| 12. Atoms (7 M) | 1 | 2 | 4 | - | 7 |
| 13*.Nuclei (10 M) | - | 2 | - | 8 | 10 |
| 14. Semiconductor Electronics (7 M) | 1 | 2 | 4 | - | 7 |
| Section wise Marks/ Total No.of Q's | 9M/9Q | 28 M/14Q | 32M/12Q | 16M/3Q | 85 /33Q |

☞ * indicates LAQ Chapters.

* *The Blue print is prepared according to the 'Model Question Paper' issued by B.I.E.*

☞ *Please, note that, at times, the public question paper gets slight deviation from the above given Blue Print.*

# CHAPTER INDEX


| | QR CODE | VSAQ P.No | SAQ P.No | LAQ P.No | OMQ P.No |
|---|---|---|---|---|---|
| 1. Electric Charges and Fields | | 62 | 32-34 | | 82 |
| 2. Electrostatic Potential | | 63-64 | 35- 39 | | |
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| 8. Electromagnetic Waves | | 73 | | | 87 |
| 9. Ray Optics | | 74 | | 21-26 | |
| 10. Wave Optics | | 75 | 49-50 | | 88 |
| 11. Dual Nature of Radiation | | 76 | 51-54 | | |
| 12. Atoms | | 77 | 55-57 | | 89 |
| 13. Nuclei | | 78 | | 27-30 | |
| 14. Semiconductor Electronics | | 79 -80 | 58-60 | | 90 |


☛ This chapter index helps faculty to revise lessons chapterwise.

☛ To get 'Synopsis & Physi Beats' of respective chapters please scan the above 'QR Code' through **Google Lens**

# SECTION INDEX

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| | | | |
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| 1. | MOVING CHARGES AND MAGNETISM | (8 Marks) | 16 - 20 |
| 2. | RAY OPTICS | (8 Marks) | 21 - 26 |
| 3. | NUCLEI | (8 Marks) | 27 - 30 |


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| | | | |
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| 10. | SEMICONDUCTOR ELECTRONICS | (4 Marks) | 58 - 60 |


## III. VSAQ CHAPTERS(61-80)


| | | | |
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| 1. | ELECTRIC CHARGES & FIELDS | (2 Marks) | 62 |
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IV. OMQ[ MCQ, FIB, OWQ] -- 81- 90

V. STAR QUESTIONS PLUS (SHINING STARS) -- 91-103

VI. 3 UNIT TESTS + 6 PAPERS (1 BMP + 5 GUESS)-- 104-118


☛ The Chapters are **Renumbered** as per VSAQ, SAQ, LAQ division.

# 30 DAYS ACTION PLAN

| DAY | CHAPTER | REMARKS |
|---|---|---|
| Day-1 | 1) ELECTRIC CHARGES & FIELDS ( 1M + 2M + 4M = 7 M) | OMQ + VSAQ |
| Day-2 | ELECTRIC CHARGES & FIELDS | SAQ |
| Day-3 | 2) ELECTRO STATIC POTENTIAL ( 2M + 4M + 4M = 10 M) | VSAQ+SAQ |
| Day-4 | ELECTRO STATIC POTENTIAL | SAQ |
| Day-5 | 3) CURRENT ELECTRICITY ( 1M + 2M + 4M = 7 M) | OMQ + VSAQ |
| Day-6 | CURRENT ELECTRICITY | SAQ |
| Day-7 | 4) MOVING CHARGES & MAGNETISM ( 2M + 8M = 10 M) | LAQ |
| Day-8 | MOVING CHARGES & MAGNETISM | VSAQ |
| Day-9 | MOVING CHARGES & MAGNETISM Total Marks in this unit : 34 Marks | LAQ + VSAQ |
| Day-10 | 5) MAGNETISM AND MATTER ( 1M + 2M + 4M = 7 M) | OMQ + VSAQ |
| Day-11 | MAGNETISM AND MATTER | SAQ |
| Day-12 | 6) ELECTRO MAGNETIC INDUCTION ( 1M + 2M + 4M = 7 M) | OMQ + VSAQ |
| Day-13 | ELECTRO MAGNETIC INDUCTION | SAQ |
| Day-14 | 7) ALTERNATING CURRENT ( 1M + 2M + 4M = 7 M) | OMQ + VSAQ |
| Day-15 | ALTERNATING CURRENT | SAQ |
| Day-16 | 8) ELECTROMAGNETIC WAVES ( 1M + 2M = 3 M) | OMQ + VSAQ |
| Day-17 | 9) RAY OPTICS ( 2M + 8M = 10 M) | LAQ |
| Day-18 | RAY OPTICS | VSAQ |
| Day-19 | RAY OPTICS Total Marks in this unit : 34 Marks | LAQ + VSAQ |
| Day-20 | 10) WAVE OPTICS ( 1M + 2M + 4M = 7 M) | OMQ + VSAQ |
| Day-21 | WAVE OPTICS | SAQ |
| Day-22 | 11) DUAL NATURE OF RADIATION ( 2M + 4M + 4M = 10 M) | VSAQ + SAQ |
| Day-23 | DUAL NATURE OF RADIATION | SAQ |
| Day-24 | 12) ATOMS ( 1M + 2M + 4M = 7 M) | OMQ + VSAQ |
| Day-25 | ATOMS | SAQ |
| Day-26 | 13) NUCLEI ( 2M + 8M = 10 M) | LAQ + VSAQ |
| Day-27 | NUCLEI | LAQ + VSAQ |
| Day-28 | NUCLEI | REVISION |
| Day-29 | 14) SEMI CONDUCTOR ELECTRONICS( 1M + 2M + 4M = 7 M) | OMQ + VSAQ |
| Day-30 | SEMI CONDUCTOR ELECTRONICS Total Marks in this unit : 41 Marks | SAQ |

☞ **30 Days** x 1 Hour Reading Time (or) **3 Days** x 10 Hours Reading Time



# BBQ - TIPS

## LAQ SCORING CHAPTERS

**Imp Chapter:** MOVING CHARGES & MAGNETISM

**Easy Chapter:** NUCLEI

**Super Scoring Pages (SSP): 16,17,21,23,27**

## SAQ SCORING CHAPTERS

**Easy Chapters :** ELECTRIC CHARGES & FIELDS, ELECTROSTATIC POTENTIAL, CURRENT ELECTRICITY, MAGNETISM AND MATTER

**Imp Chapters:** ELECTROMAGNETIC INDUCTION, DUAL NATURE, ATOMS

**Super Scoring Pages (SSP): 32, 33, 35, 36,37, 40, 43, 45, 46, 48, 49, 51, 52, 54, 55,59**

## VSAQ SCORING CHAPTERS

**Easy Chapters:** DUAL NATURE, ALTERNATING CURRENT, ELECTRO MAGNETIC WAVES, SEMICONDUCTORS,

**Imp Chapters:** ELECTRIC CHARGES, ELECTROSTATIC POTENTIAL, MOVING CHARGES, ATOMS, NUCLEI

**Super Scoring Pages (SSP):62, 63, 65, 67, 68, 69, 71, 73, 74, 75, 76, 77, 78, 79**

## BBQ HIGHLIGHTS

- **Quick Memory Bytes → (i) Answer Clarity (ii) Easy Memory .**
- Chapter concept పై Motivation కావాలంటే **Physi beats** మీ కోసం రెడీ! ( on QR Code).
- **Rising Stars** కోసం **Level I** Questions, **Shining Stars** కోసం **Level II** Questions (Star Q Plus).
- **30 Days Action Plan : 30 Days** x 1 Hour Reading Time.
- **3 Days Action Plan during Public Exam : 3 Days** x 10 Hours Reading Time.
- **3 Unit Tests to complete the entire Physics syllabus in 3 units.**

☞ **SINCERE SUGGESTION:** After completing this BulletQ, Shining stars are advised to go through the Notes given by your Lecturers; follow the next Higher Level Material.

# IMP FORMULAS FOR IPE PROBLEMS

## ELECTROSTATIC POTENTIAL

- Potential due to a point charge is $V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$
- Capacitance of a capacitor $C = \frac{Q}{V}$
- Capacitance of a parallel plate capacitor is $C = \frac{\varepsilon_0 A}{d}$
- Equivalent capacitance of 2 capacitors in parallel is $C = C_1 + C_2$
- Equivalent capacitance of 2 capacitors in series is given by $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$
- Energy stored in a capacitor is $U = \frac{Q^2}{2C}$ or $U = \frac{1}{2}QV$ or $U = \frac{1}{2}CV^2$

## MOVING CHARGES & MAGNETISM

- The magnetic force on a charged particle moving in a magnetic field is $F = qvB\sin\theta$
- Magnetic field $B_a = \frac{\mu_0 I_a}{2\pi d}$
- Force $F_{ba} = B_a I_b l$
- Force per unit length $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi r}$
- Torque = $BIAN\sin\theta$
- Maximum torque, $\tau_{max} = BIAN$
- Shunt Resistance $r_s = \frac{I_G R_G}{I - I_G}$
- Series Resistance $R = \frac{V}{I_G} - R_G$

## RAY OPTICS

- **Snell's law of refraction** :
  Refractive index $n_{21} = \frac{\sin i}{\sin r}$ = constant
  (or) $n_1 \sin i = n_2 \sin r$
- **Mirror formula:** $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$
- Combined focal length (f) of thin lenses in contact: $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$
- Power (P) of a lens: $P = \frac{1}{f}$
- Refractive index of material of the prism
$$n_{21} = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

---

For a thin prism, $D_m = (n_{21} - 1)A$

## DUAL NATURE OF RADIATION

- Threshold frequency $\nu_0 = \phi_0 / h$
- Einstein's photoelectirc equation $eV_0 = h\nu - \phi$
- Wavelength $\lambda = \frac{hc}{(eV_0 + \phi_0)}$
- Threshold wavelength $\lambda_0 = \frac{hc}{\phi}$
- de-Broglie wave length $\lambda = \frac{h}{p}$
- K.E of electron $K = \frac{h^2}{2m\lambda^2}$
- Einstein's photoelectric equation is
$K_{max} = h\nu - \phi_0$ or $\frac{1}{2}mv^2_{max} = h(\nu - \nu_0)$

## NUCLEI

- Radius of nucleus $R = R_0 A^{1/3}$
- Mass defect $\Delta M = [Zm_P + (A-Z)m_n] - M$
- Energy $E = mc^2$
- **Nuclear fission:**
$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^{1}_{0}n + 200MeV$
- **Nuclear fusion:**
$4\,{}^{1}_{1}H + 4\,{}_{-1}e^{0} \rightarrow {}^{4}_{2}He + 2\,{}_{-1}e^{0} + 2\nu + 6\gamma + 26.7MeV$

## SEMI CONDUCTORS

- Half wave rectifier efficiency $\eta = \frac{0.406R_L}{r_f + R_L}$
- Full wave rectifier efficiency $\eta = \frac{0.812R_L}{r_f + R_L}$
- Current gain in a transistor is $\beta = \frac{\Delta I_C}{\Delta I_B}$



# ★ 8 MARKS LAQ HIT LIST

## TOP 10


| | | Ans-Page |
|---|---|---|
| | ★★★ MOVING CHARGES & MAGENTISM [ 8 MARKS ] ★★★ | |
| LAQ 1. | Derive an expression for the force acting between two very long parallel current carrying conductors and hence define the ampere with a neat diagram? | [☞P 16(1)] |
| LAQ 2. | Explain how can a Moving coil galvanometer be converted into (a) Ammeter (b)Voltmeter with a neat diagram. | [☞P 17(2)] |
| LAQ 3. | Obtain an expression for the torque on a current carrying loop placed in a uniform magnetic field. Describe the construction and working of a moving coil galvanometer with a neat diagram. | [☞P 18(3)] |
| LAQ 4. | State and explain Biot-Savart law, thereby derive an expression for magnetic field induction at any point on the axis of circular current carrying coil with neat diagrams. | [☞P 19(4)] |
| | ★★ RAY OPTICS [ 8 MARKS ] ★★ | |
| LAQ 5. | Derive the mirror equation and magnification formula for a concave mirror in case of object placed beyond the center of curvature with a neat ray diagram. | [☞P 21(5)] |
| LAQ 6. | a)Explain the total internal reflection of light with a neat ray diagram. Derive the relation between critical angle and refractive index.<br>b)Explain the working of optical fiber with a neat diagram. | [☞P 22(6)] |
| LAQ 7. | Derive the equation for angle of deviation produced by a triangular prism with a neat labelled diagram, and there by derive an expression for refractive index of material of the prism. | [☞P 23(7)] |
| | ★★★ NUCLEI [ 8 MARKS ] ★★★ | |
| LAQ 8. | Explain nuclear fission and fusion with suitable nuclear reactions. | [☞P 27(10)] |
| LAQ 9. | a)What is the source of stellar energy? Explain it with the help of proton-proton cycle.<br>b)Briefly explain the formation of red giant stage of sun. | [☞P 27(11)] |
| LAQ 10. | Define mass defect and binding energy. How does binding energy per nucleon vary with mass number? What is its significance? | [☞P 29(13)] |


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# ★ 4 MARKS SAQ HIT LIST

## TOP 40

### ★★★ELECTRIC CHARGES & FIELDS [4 MARKS] ★★★

| | | Ans-Page |
|---|---|---|
| SAQ 1. | State and explain Coulomb's inverse square law in electricity. | [☞P 32(14)] |
| SAQ 2. | Define intensity of electric field at a point. Derive an expression for the intensity due to a point charge. | [☞P 32(15)] |
| SAQ 3. | Derive an equation for the couple acting on an electric dipole in a uniform electric field. | [☞P 33(16)] |
| SAQ 4. | Derive an expression for the intensity of the electric field at a point on the axial line of a dipole. | [☞P 33(17)] |
| SAQ 5. | Derive an expression for the intensity of the electric field at a point on the equatorial plane of an electric dipole. | [☞P 34(18)] |
| SAQ 6. | State Gauss's law in electrostatics and prove it. | [☞P 34(19)] |

### ★★★ELECTROSTATIC POTENTIAL [4+4 MARKS] ★★★

| | | |
|---|---|---|
| SAQ 7. | Derive an expression for the electric potential due to a point charge. | [☞P 35(20)] |
| SAQ 8. | Derive an expression for the potential energy of an electric dipole placed in a uniform electric field. | [☞P 36(22)] |
| SAQ 9. | Derive an expression for the capacitance of a parallel plate capacitor. | [☞P 36(23)] |
| SAQ 10. | Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in series. | [☞P 37(24)] |
| SAQ 11. | Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in parallel. | [☞P 37(25)] |

### ★★★CURRENT ELECTRICITY [4 MARKS] ★★★

| | | |
|---|---|---|
| SAQ 12. | Using Kirchhoff's rules deduce the condition for balance in a Wheatstone bridge. | [☞P 40(32)] |
| SAQ 13. | Derive equivalent emf and internal resistance of two cells in series combination. | [☞P 41(33)] |
| SAQ 14. | Derive equivalent emf and internal resistance of two cells in parallel combination. | [☞P 42(36)] |
| SAQ 15. | State and explain Kirchhoff rules for an electric network. | [☞P 41(34)] |

### ★★★MAGNETISM & MATTER [4 MARKS] ★★★

| | | |
|---|---|---|
| SAQ 16. | Mention the properties of magnetic field lines. | [☞P 43(37)] |
| SAQ 17. | Explain the properties of diamagnetic materials. | [☞P 43(38)] |
| SAQ 18. | Explain the properties of paramagnetic materials. | [☞P 43(39)] |
| SAQ 19. | How the existence of domains, explain the behavior of ferromagnetic material? | [☞P 44(40)] |

### ★★ELECTRO MAGNETIC INDUCTION [4 MARKS] ★★

| | | |
|---|---|---|
| SAQ 20. | Explain the Faradays law of induction and Lenz's law. | [☞P 45(43)] |
| SAQ 21. | Derive an expression for motional emf induced across the conductor moving perpendicular to uniform magnetic field. | [☞P 45(44)] |
| SAQ 22. | Explain briefly working of AC generator. | [☞P 46(45)] |
| SAQ 23. | Obtain an expression for the magnetic energy stored in a solenoid in terms of the magnetic field, area and length of the solenoid. | [☞P 46(46)] |



### ★★ ALTERNATING CURRENT[4 MARKS] ★★

| | | |
|---|---|---|
| SAQ 24. | State the principle on which a transformer works. Describe the working of a transformer with necessary theory. | [☞P 47(47)] |
| SAQ 25. | Obtain an expression for the current through an inductor when an AC emf is applied. | [☞P 48(49)] |
| SAQ 26. | Obtain an expression for the current in a capacitor when an AC emf is applied. | [☞P 48(50)] |

### ★★ WAVE OPTICS [4 MARKS] ★★

| | | |
|---|---|---|
| SAQ 27. | Derive an expression for the intensity at a point where interference of light occurs. Arrive the conditions for maximum and zero intensity. | [☞P 49(51)] |
| SAQ 28. | Does principle of conservation of energy hold for interference and diffraction phenomenon? Explain briefly. | [☞P 49(52)] |
| SAQ 29. | Discuss the intensity of light when a polaroid sheet is rotated between two crossed polaroids. | [☞P 50(54)] |

### ★★★ DUAL NATURE OF RADIATION [4+4 MARKS] ★★★

| | | |
|---|---|---|
| SAQ 30. | What are different types of physical processes to obtain free electrons from the metal surface. | [☞P 51(55)] |
| SAQ 31. | Define (i) Threshold frequency (ii) Work function and give the relation between them and define stopping potential. | [☞P 51(56)] |
| SAQ 32. | What are the laws of photoelectric effect. | [☞P 52(57)] |
| SAQ 33. | The work function of caesium is 2.14 eV. Find (a) the threshold frequency for caesium, and (b) the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.60 V. | [☞P 54(61)] |
| SAQ 34. | What is the kinetic energy of an electron whose de-Broglie wavelength is 5000Å ? | [☞P 54(64)] |

### ★★★ ATOMS [4 MARKS] ★★★

| | | |
|---|---|---|
| SAQ 35. | State Bohr's postulates of hydrogen atom. | [☞P 55(65)] |
| SAQ 36. | What are the limitations of Bohr's theory of hydrogen atom? | [☞P 55(66)] |
| SAQ 37. | Write a short note on debroglie's explanation of Bohr's second postulate of quantization? | [☞P 56(68)] |

### ★★ SEMI CONDUCTORS [4 MARKS] ★★

| | | |
|---|---|---|
| SAQ 38. | What are n-type and p-type semiconductors? Explain the process of formation of p-n junction diode. | [☞P 58(71)] |
| SAQ 39. | Describe how a semiconductor diode is used as a half wave rectifier. | [☞P 59(72)] |
| SAQ 40. | What is rectification? Explain the working of a full wave rectifier. | [☞P 59(73)] |

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# ★ 2 MARKS VSAQ HIT LIST

## TOP 70


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| VSAQ 1. | The electric lines of force do not intersect. Why ? | [☞P 62(78)] |
| VSAQ 2. | What is meant by the statement charge is quantised ? | [☞P 62(79)] |
| VSAQ 3. | How many electrons constitute 1C of charge ? | [☞P 62(80)] |
| VSAQ 4. | What is the importance of Gauss law in electrostatics. | [☞P 62(81)] |
| VSAQ 5. | What happens to the force between two charges if the distance between them is a) halved b) doubled | [☞P 91(182)] |
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| VSAQ 6. | Can there be electric potential at a point with zero electric intensity? Give an example. | [☞P 63(84)] |
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| VSAQ 8. | Three capacitors of capacitances 1μF, 2μF and 3μF are connected in parallel. a)What is the ratio of charges on each capacitor? b)What is the ratio of potential differences across each capacitor? | [☞P 63(88)] |
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| VSAQ 10. | What happens to the capacitance of a parallel plate capacitor if the area of its plates is doubled? | [☞P 64(90)] |
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| VSAQ 13. | Write the vector form of Ohms law. Explain the terms in it. | [☞P 65(95)] |
| VSAQ 14. | Define temperature coefficient of resistance. | [☞P 65(97)] |
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| | | |
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| | ★★★ MAGNETISM & MATTER [2 MARKS] ★★★ | |
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| VSAQ 32. | What is transformer ratio? | [☞P 71(125)] |
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| VSAQ 42. | What is the wavelength range of X-rays? | [☞P 73(141)] |
| VSAQ 43. | If the wavelength of electromagnetic radiation is doubled, what happens to the energy of photon? | [☞P 94(197)] |

### ★★☆ RAY OPTICS [2 MARKS] ★★☆

VSAQ 44. Define power of a convex lens. What is its unit? [☞P 74(142)]
VSAQ 45. Two thin lens of power +4D and -2D are in contact. What is the focal length of the combination? [☞P 74(143)]
VSAQ 46. What is the critical angle? Give one application of total internal reflection? [☞P 74(145)]
VSAQ 47. A small angled prism of 4° deviates a ray through 2.4°. Find the refractive index of the prism. [☞P 74(147)]

### ★★☆ WAVE OPTICS [2 MARKS] ★★☆

VSAQ 48. State Malus' law. Write the expression of Malus' law. [☞P 75(148)]
VSAQ 49. What is polarization of light? Which field vector, electric or magnetic, is used to represent the polarization of an electromagnetic waves? [☞P 75(149)]
VSAQ 50. If a wave undergoes reflection and refraction, what will be the phase change in each case? [☞P 75(151)]
VSAQ 51. Compare Maxwell's Electromagnetic theory with Huygen's theory of light? [☞P 75(152)]
VSAQ 52. Explain the statement *'light added to light can produce darkness'* [☞P 97(209)]

### ★★★ DUAL NATURE [2 MARKS] ★★★

VSAQ 53. What are cathode rays? [☞P 76(153)]
VSAQ 54. What is photoelectric effect? [☞P 76(155)]
VSAQ 55. Write down Einstein's photoelectric equation. Explain the terms in it? [☞P 76(158)]
VSAQ 56. Write down de-Broglie's relation and explain the terms therein. [☞P 76(159)]
VSAQ 57. Define 1 electron volt. [☞P 76(160)]

### ★★☆ ATOMS [2 MARKS] ★★☆

VSAQ 58. Define impact parameter. [☞P 77(162)]
VSAQ 59. What is the difference between α - particle and helium atom? [☞P 77(164)]
VSAQ 60. Why did most of the α - particles pass through the gold foil undeflected in Rutherford's scattering? [☞P 77(165)]

### ★★★ NUCLEI [2 MARKS] ★★★

VSAQ 61. What are isotopes and isobars? [☞P 78(168)]
VSAQ 62. What are isotones and isomers? [☞P 78(169)]
VSAQ 63. Why are nuclear fusion reactions called thermo nuclear reactions? [☞P 78(170)]
VSAQ 64. Compare the radii of the nuclei of mass numbers 27 and 64. [☞P 78(171)]
VSAQ 65. What is the significance of a mass defect in a nucleus? [☞P 78(172)]

### ★★★ SEMI CONDUCTORS[2 MARKS] ★★★

VSAQ 66. What are intrinsic and extrinsic semiconductors ? [☞P 79(173)]
VSAQ 67. What is an n-type semiconductor? What are the majority and minority charge carriers in it? [☞P 79(174)]
VSAQ 68. What is a p-type semiconductor ? What are the majority and minority charge carriers in it? [☞P 79(175)]
VSAQ 69. How is a battery connected to a junction diode in (1) forward and (2) reverse bias? [☞P 79(176)]
VSAQ 70. What happens to the width of the depletion layer in a p-n junction diode when it is i) forward-biased and ii) reverse biased? [☞P 80(180)]



RISING STARS' MOST FAVOURITE SECTION

# LAQ

## SECTION-D

BBQ turns, "I can't to I can".

Small book, Big Marks.

## LAQ CHAPTERS

16 MARKS - 12 PAGES

| | | |
|---|---|---|
| ★★★ 1. | MOVING CHARGES & MAGNETISM (8 Marks) | 16 - 20 |
| ★★ 2. | RAY OPTICS (8 Marks) | 21 - 26 |
| ★★★ 3. | NUCLEI (8 Marks) | 27 - 30 |

**WOW TIP 1:** First chapter is a 'must and should' read chapter. It is better to practice all the 4 LAQs.

**WOW TIP 2:** In Ray Optics Q5, Q6 and also Q7 are easy and important.

**WOW TIP 3:** Q10 & Q11 are the easiest of all LAQs. Practice all the 3 tag problems of Nuclei.

☞ Practice the **diagrams** perfectly in these LAQs, as they consume little more time in the exam.

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**Super Scoring Pages**

★★★SSP★★★

**Page 16,17,21,23,27**

**Total SS Pages : 5**

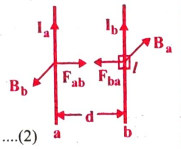
# 1 MOVING CHARGES & MAGNETISM

**8 Marks**

**1. Derive an expression for the force acting between two very long parallel current carrying conductors and hence define the ampere with a neat diagram? [BMP]**

**A: (a) Force between two parallel conductors carrying Current:**

1) Consider two straight parallel conductors a, b carrying currents $I_a$, $I_b$ separated by a distance d .




2) Magnetic field produced by conductor a is $B_a = \frac{\mu_0 I_a}{2\pi d}$ .....(1)

3) Force on a segment $l$ of b by magnetic field $B_a$ is $F_{ba}=B_a I_b l$ ....(2)

From (1) & (2), $F_{ba} = \left(\frac{\mu_0 I_a}{2\pi d}\right) I_b l = \frac{\mu_0 I_a I_b l}{2\pi d}$.....(3)

The direction of this force $F_{ba}$ is towards conductor a.

4) Similarly, the force acting on the conductor a by magnetic field $B_b$ is $F_{ab}=\frac{\mu_0 I_a I_b l}{2\pi d}$ ....(4)

The direction of this force $F_{ab}$ is towards conductor b.

5) From (3) & (4), we get $F_{ab}= F_{ba}=\frac{\mu_0 I_a I_b l}{2\pi d}$ ....(5)

From (5), force per unit length is $\frac{F}{l}=\frac{\mu_0 I_a I_b}{2\pi d}$

**(b) Ampere:** One ampere is the value of steady current when two infinitely long parallel conductors, carrying the same current separated by a distance of 1m in vacuum produces a force of $2\times10^{-7}$ $Nm^{-1}$ on each conductor.

**QUICK MEMORY BYTES**

1) Imagine two friends a,b ( wires) walking side by side .
2) Friend a creates a magnetic field $B_a$ around him.
3) Friend b feels a force of attraction $F_{ba}$ due to a.
4) Similarly, friend a also feels a force of attraction $F_{ab}$ due to b.
5) Infact, the two forces are equal i.e $F_{ba}= F_{ab}$

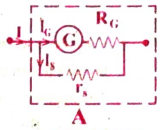
**Formula Ladder**

- $B_a = \frac{\mu_0 I_a}{2\pi d}$
- $F_{ba}=B_a I_b l$
- $\frac{F}{l}=\frac{\mu_0 I_a I_b}{2\pi d}$



**2. Explain how can a Moving coil galvanometer be converted into**
**(a) Ammeter (b)Voltmeter with a neat diagram.**

**A: (a) Conversion of Galvanometer into Ammeter:**




(a) Ammeter

1) Galvanometer (G) measures only small currents.
2) To measure a large current (I), a galvanometer (G) is converted into an ammeter (A) by connecting a very small shunt resistance ($r_s$) parallel to it.
3) Then current splits and most of the current passes through $r_S$.

**Derivation of Shunt Resistance ($r_S$) :**

4) From the Kirchoff's 1st law, max. current $I= I_G+I_S \Rightarrow I_S= I-I_G$
5) Galvanometer and shunt are in parallel, so the potential difference across them is same:

$\therefore I_G R_G=I_S r_S \Rightarrow I_G R_G=(I-I_G) r_S \Rightarrow r_S = \frac{I_G R_G}{I-I_G}$

This gives the value of 'shunt resistance' required to convert galvanometer into an ammeter.

**(b) Conversion of Galvanometer into Voltmeter:**




(b) Voltmeter

1) Galvanometer (G) measures only small voltages.
2) To measure a large voltage (V), a galvanometer (G) can be converted into a voltmeter by connecting a high resistance (R) in series with it.
3) This high resistance limits the current and makes the total resistance of the voltmeter very large.

**Derivation of Series Resistance (R):**

4) Resistance of G is $R_G$ and current through G is $I_G$ ( at full scale deflection)
5) **From Ohm's law:** Voltage $V=I_G(R_G+R) \Rightarrow R_G + R = \frac{V}{I_G} \Rightarrow R = \frac{V}{I_G} - R_G$ ,

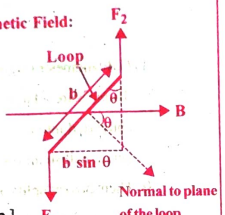
This gives the value of 'high resistance' required to convert galvanometer into an voltmeter.

**QUICK MEMORY BYTES**

**GA→ SR P**
Galvanometer to Ammeter → Small shunt Resistance in Parallel
**GV→ HR S**
Galvanometer to Voltmeter → High Resistance in Series

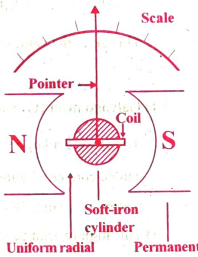
**3. Obtain an expression for the torque on a current carrying loop placed in a uniform magnetic field. Describe the construction and working of a moving coil galvanometer with a neat diagram.**

**A: (a) Torque on a current carrying Loop in a Uniform Magnetic Field:**




1) Consider a rectangular current loop of length $l$, breadth b is suspended in a magnetic field B.
2) The magnetic forces on the arms are given by $F_1 = F_2 = Bil$
3) The two forces constitute a couple.
4) The perpendicular distance between the forces is $b\sin\theta$. [$\theta$ is the angle between the field and the normal to the loop.]
5) Torque = Force × Perpendicular distance
$\tau = Bil \times b\sin\theta = B\,i\,A\sin\theta$ [$\because$ Area of the loop $lb = A$]

**(b) Construction and working of MCG:**
The current-carrying coil placed in uniform magnetic field experience a torque.




**Moving coil galvanometer**

**Construction:**
1) MCG consists of a rectangular coil with a pivot arrangement in a strong radial magnetic field NS.
2) A 'soft iron cylinder' is arranged at the 'centre of the coil' to produce the uniform magnetic field.
3) A spring is arranged to the lower end of the coil.

**Working:** If the current I to be measured is passed through the MCG, the coil gets a deflection $\theta$.
Deflecting torque acting on the coil is $\tau = BIAN$ ....(1)
Restoring torque developed in the suspension is $\tau_r = C\theta$....(2)
Here C is the torsional torque.
At equilibrium position, the deflecting magnetic torque = restoring torque due to spring

From (1) & (2), $BIAN = C\theta \Rightarrow I = \frac{C\theta}{BAN}$. Put $\frac{C}{BAN} = K$

Hence, $I = K\theta \Rightarrow I \propto \theta$ Here K is called the constant of the galvanometer.

The relation $I = K\theta$ explains the principle of MCG.

By determining the values of K and $\theta$, the current through MCG can be calculated.

**FORMULA BOX**

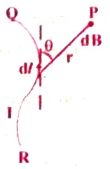
- $F_1 = F_2 = Bil$, $\tau = B\,i\,A\sin\theta$
- $BIAN = C\theta \Rightarrow I = \frac{C\theta}{BAN}$; $\frac{C}{BAN} = K \Rightarrow I = K\theta$



**4. State and explain Biot-Savart law, thereby derive an expression for magnetic field induction at any point on the axis of circular current carrying coil with neat diagrams.**

**A: Biot-Savart Law:** The magnitude of magnetic field induction due to small element of current carrying conductor is (i) directly proportional to the strength of the current (ii) length of the element (iii) sine of the angle between position vector and the element and (iv) inversely proportional to the square of the distance of the point from the element.

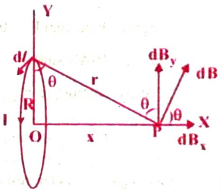
**Explanation:** Suppose RQ is a conductor carrying a current I.
$dl$ is a small length of element on the conductor.
The distance of a point P from $dl$ is r.
Let $\theta$ be the angle between $\overline{dl}$ and radius vector $\overline{r}$.
The magnetic induction due to $dl$ is dB.
From Biot-savart law, we have




(i) $dB \propto I$ (ii) $dB \propto dl$ (iii) $dB \propto \sin\theta$ (iv) $dB \propto \frac{1}{r^2}$

Thus, $dB \propto \frac{Idl\sin\theta}{r^2}$ (or) $dB = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$ Here $\frac{\mu_0}{4\pi} = 10^{-7}\ Hm^{-1}$

**Magnetic Induction on the axis a coil:**




1) Consider a circular loop of radius R carrying a current I.
2) Let P be a point on the OX- axis at a distance x from its centre O.
3) Consider a small element $dl$ carrying the current I on the coil.
4) The distance of P from the element $dl$ is r.
5) From the Biot-Savart law, $dB = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$, [dB is the magnetic induction at P due to $dl$ ]
$\therefore dB = \frac{\mu_0}{4\pi}\frac{Idl}{r^2}$ [$\because$ Angle between $dl$ and r is 90°.]

$\Rightarrow dB = \frac{\mu_0}{4\pi}\frac{Idl}{(x^2+R^2)}$ [From the figure, $r^2 = x^2 + R^2$ ]

$\Rightarrow B = \int_0^{2\pi R} dB\cos\theta = \frac{\mu_0}{4\pi}\frac{I}{(x^2+R^2)}\cos\theta\int_0^{2\pi R} dl$ [B is the resultant M.I at 'P' due to circular loop]

6) From the figure, $\cos\theta = \frac{R}{r} = \frac{R}{(x^2+R^2)^{1/2}}$ & for the circular coil $\int_0^{2\pi R} dl = 2\pi R$

$\therefore B = \frac{\mu_0}{4\pi}\frac{I}{(x^2+R^2)}\frac{R}{(x^2+R^2)^{1/2}}\times 2\pi R \Rightarrow B = \frac{\mu_0 IR^2}{2(x^2+R^2)^{3/2}}$

If the coil has N turns, magnetic filed induction $B = \frac{\mu_0 NIR^2}{2(x^2+R^2)^{3/2}}$

## TAG PROBLEMS

**P 1. A current of 5A flows through each of two parallel long wires, the wires are 2.5 cms apart. Calculate the force acting per unit length of each wire. [BMP]**

**Sol:** Current in each wire $I_1 = I_2 = 5A$

Distance between the wires r = 2.5 cm = 0.025 m. Also $\mu_0 = 4\pi\times10^{-7}$

Force per unit length $\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi r} = \frac{[4\pi\times10^{-7}][5][5]}{2\pi(0.025)} = 2\times10^{-4}$ Nm$^{-1}$

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**P 2. A coil of area $10^{-2}m^2$ and contains $10^6$ turns is used in a galvanometer. What is the maximum torque on it if the galvanometer carrying a current of $10^{-4}$ A is placed in a magnetic field of $10^2$ T?**

**Sol:** Given area of the coil A = $10^{-2}m^2$, Number of turns N = $10^6$, Current I= $10^{-4}$ A

Magnetic induction B =$10^2$ T, Torque will be maximum when $\theta = 90° \Rightarrow \sin 90°=1$

Torque = BIAN sinθ

$\therefore$ Maximum torque, $\tau_{max}$ = BIAN = $10^2\times10^{-4}\times10^{-2}\times10^6$ = 100Nm

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**P 3. A Moving coil galvanometer, having a resistance of 1 ohm can measure a current $10^{-6}$ A. What is the resistance of shunt required to measure a current 1A ?**

**Sol:** Given galvanometer, resistance $R_G$ =1Ω, Current through galvanometer $I_G = 10^{-6}$ A,

Current to be measured, I= 1A, $r_s$ = ?

Shunt Resistance $r_s = \frac{I_G R_G}{I - I_G} = \frac{10^{-6}\times1}{1-10^{-6}} \approx 10^{-6}\Omega$ $(\because 1-10^{-6}\approx1)$

☞ Few more **Tag Problems** suggested by your **Faculty.**

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# 2 RAY OPTICS

**8 Marks**

**5. Derive the mirror equation and magnification formula for a concave mirror in case of object placed beyond the center of curvature with a neat ray diagram.**

**A: Derivation of Mirror Equation :**

1) Consider a concave mirror with pole P, focus F, and centre of curvature C.

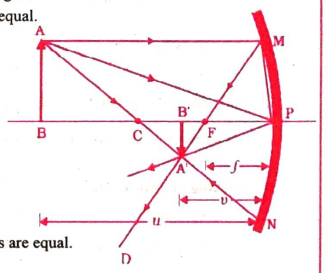
2) AB is an object in front of the mirror. A'B' is its image formed by the mirror.

PF = f = focal length of concave mirror; PB' = v = image distance; PB = u = object distance.

3) In the figure, Δ FB'A', ΔMFP are similar triangles.

So, the ratio of their corresponding sides are equal.

$\therefore \frac{B'A'}{MP} = \frac{B'F}{FP}$




4) But from the figure, MP = AB

$\therefore \frac{B'A'}{AB} = \frac{B'F}{FP}$ .....(1)

5) Δ A'B'P, ΔABP are also similar triangles.

Hence, the ratio of their corresponding sides are equal.

$\therefore \frac{B'A'}{AB} = \frac{B'P}{BP}$ .....(2)

6) From (1) and (2), $\frac{B'F}{FP} = \frac{B'P}{BP}$

7) But from the figure, B'F = B'P–FP $\therefore \frac{B'P-FP}{FP} = \frac{B'P}{BP}$ .....(3)

8) From sign convention, distances from P against the direction of incident light are negative.

Here B'P = –v, FP = –f, BP = –u

9) From (3), we get $\frac{-v-(-f)}{-f} = \frac{-v}{-u} \Rightarrow \frac{v-f}{f} = \frac{v}{u} \Rightarrow \frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

This is the **mirror equation.**

10) **Linear Magnification (m) :**

It is the ratio of height of the image (h') to height of the object (h) $\therefore m = \frac{h'}{h}$

In the case of a mirror with proper sign convention, we have $m = -\frac{v}{u}$

Hence, we conclude that a **real inverted image** is formed by a concave mirror.

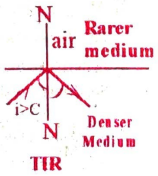
**6. a) Explain the total internal reflection of light with a neat ray diagram. Derive the relation between critical angle and refractive index.**

**b) Explain the working of optical fiber with a neat diagram.**

**A: a) Total Internal Reflection (TIR):**

1) When a light ray is passing from denser medium to rarer medium and if the angle of incidence is greater than its critical angle ($i > i_C$), it gets reflected into the same denser medium without any refraction. This is called total internal reflection.

**Relation between critical angle and referactive index:**




2) Let $n_2$ be the refractive index of the denser medium
$n_1$ be the refractive index of the rarer medium .

3) From Snell's law, $\frac{n_2}{n_1} = \frac{\sin r}{\sin i_C} = \frac{1}{\sin i_C}$ $(r = 90° \Rightarrow \sin 90° = 1)$

4) Equation for critical angle is $n_{21} = \frac{1}{\sin i_C}$ (or) $\sin i_C = \frac{1}{n_{21}}$

where $n_{21}$ = refractive index of the denser medium with respect to rarer medium.

**b) Working of optical fiber:**



Optical fibre

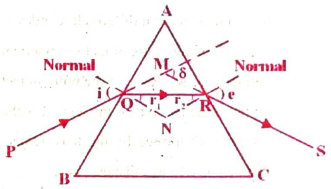
1) An optical fibre works on the principle of **TIR.**
2) Optical fibre consists of a **core and cladding.** The refractive index of the material of the core is higher than that of the cladding.
3) When a light signal is directed at one end of the fibre at a suitable angle, it undergoes repeated total internal reflections along the length of the fibre and finally comes out at the other end.
4) Since light undergoes TIR at each stage, there is no appreciable loss in the intensity of the light signal.
5) Optical fibres are fabricated such that light reflected at one side of inner surface strikes the other at an angle larger than the critical angle. Even if the fibre is bent, then also light can easily travel along its length.

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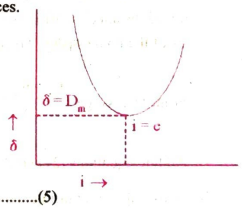


**7. Derive the equation for angle of deviation produced by a triangular prism with a neat labelled diagram, and there by derive an expression for refractive index of material of the prism. [BMP]**

**A: (a) Angle of incidence (i) , Angle of minimum deviation ($D_m$):**




1) Consider a triangular glass prism ABC.
A = Refracting angle of the prism.
$n_{21}$ = Refractive index of glass w.r.to air.
PQ = Incident ray.
QR = Refracted ray in the prism.
RS = Emergent ray
2) The normals to AB and AC meet at N.
i = Angle of incidence at the first surface.
$r_1$ = Angle of refraction at the first surface.
$r_2$ = Angle of refraction at the second surface
e = Angle of emergence.
δ = Angle of deviation ( angle between PQ and RS).
3) In the quadrilateral AQNR , $A + 90° + N + 90° = 360° \Rightarrow A + N = 180°$ .....(1)
4) In Δ QNR, $r_1 + r_2 + N = 180°$ .....(2)
5) Equating eq. (1) and (2), $r_1 + r_2 + N = A + N \Rightarrow r_1 + r_2 = A$ .....(3)
6) The total deviation δ = sum of deviations at the two faces.
$\therefore \delta = (i - r_1) + (e - r_2)$
$\Rightarrow \delta = i + e - (r_1 + r_2) = i + e - A$ [From (3)]
$\Rightarrow \delta + A = i + e$ .....(4)
7) A graph is drawn between angle of incidence i on the x-axis and angle of deviation δ on the y-axis.
8) From graph, we get $i = e$, $r_1 = r_2 = r$, $\delta = D_m$




From (3) $r_1 + r_2 = A \Rightarrow r + r = A \Rightarrow 2r = A \Rightarrow r = \left(\frac{A}{2}\right)$ .....(5)

From(4) $\delta + A = i + e \Rightarrow D_m + A = i + i \Rightarrow 2i = A + D_m \Rightarrow i = \left(\frac{A + D_m}{2}\right)$.....(6)

This is the equation for angle of incidence (i) and also angle of minimum deviation ($D_m$).

**(b) Prism formula for refractive index ($n_{21}$) of prism:**

When the light ray is passing from medium 1 (air) to medium 2 (glass), the refractive index ($n_{21}$) of medium 2 with respect to medium 1 is given by $n_{21} = \frac{\sin i}{\sin r}$ .....(7)

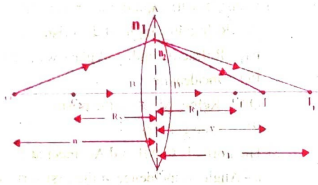
From (5) & (6), we get $n_{21} = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

**8. Derive the lens makers formula with suitable diagram and thereby derive the lens formula.**

**A: Lens makers formula & Thin lens formula:**

1) Consider a thin double convex lens of material of refractive index $n_2$, placed in a medium of refractive index $n_1$.

2) The radii of curvature of its two spherical surfaces are $R_1$, $R_2$. O is the point object on the principal axis of the lens . I is the image of O formed by the lens. Object distance is u Image distance is v.



3) The general formula of refraction for curved surface is $\frac{n_2}{v}-\frac{n_1}{u}=\frac{n_2-n_1}{R}$ .....(1)

**4) For first surface:** Object distance =u, image distance =v, RI of the object space =$n_1$, RI of the image space =$n_2$, Radius of curvature = $R_1$

From (1), we get $\frac{n_2}{v_1}-\frac{n_1}{u}=\frac{n_2-n_1}{R_1}$ .....(2)

**5) For second surface:** Object distance =$v_1$, image distance =v, RI of the object space =$n_2$, RI of the image space =$n_1$, Radius of curvature = $R_2$

From (1), we get $\frac{n_1}{v}-\frac{n_2}{v_1}=\frac{n_2-n_1}{-R_2}$ .....(3)

Adding equations (2) and (3), we get $\frac{n_1}{v}-\frac{n_1}{u}=(n_2-n_1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$ .....(4)

6) If the object is at infinity (u = ∞), its image is formed at focus (v = f).

Putting these in equ (4), we get $\frac{n_1}{f}=(n_2-n_1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$ .....(5)

Dividing it by $n_1$, we get $\frac{1}{f}=(n_{21}-1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$ ---(5) where $n_{21}=\frac{n_2}{n_1}$

This is the **Lens maker's formula.**

By comparing (4) and (5), we get $\frac{n_1}{v}-\frac{n_1}{u}=\frac{n_1}{f} \Rightarrow \frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ .....(6)

This is the **Thin lens formula.**



**9. With a neat labelled diagram explain the formation of image in a simple microscope. Derive the expression for magnification when the image is formed at**

**i) near point ii) at infinity.**

**A: Simple Microscope:**

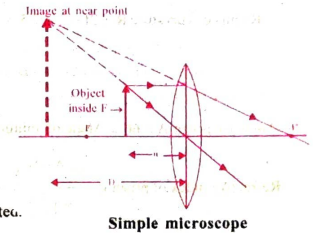
1) A simple microscope or magnifying glass is a convex lens of small focal length.

2) In this, the object is placed between the principal focus F and the optic centre of the convex lens.

3) The light ray coming from the object parallel to principal axis refracts at lens and passes through second principal focus F'.

4) Another light ray from the object passes through the optic centre of the lens undeviated.

5) These two rays meet on the back side of the object to form the image.




**Simple microscope image at near point**

Hence, virtual, erect and magnified image is formed on the same side of object at D = 25 cm.

**6) Expression for Magnification :** It is the ratio of distance of image to the distance of object.

$$m=\frac{\text{Distance of image}}{\text{Distance of object}}=\frac{D}{u}$$

**7) From Len's formula,** $\frac{1}{f}=\frac{1}{v}-\frac{1}{u} \Rightarrow \frac{1}{f}=-\frac{1}{v}+\frac{1}{u} \Rightarrow \frac{1}{u}=\frac{1}{v}+\frac{1}{f}$

$$\therefore m=D\left(\frac{1}{v}+\frac{1}{f}\right)=\frac{D}{v}+\frac{D}{f}$$

**i) At near point, v=D**

$$\therefore m=\frac{D}{v}+\frac{D}{f}=\frac{D}{D}+\frac{D}{f}=1+\frac{D}{f}$$

**ii) At far point , v=∞**

$$\therefore m=\frac{D}{v}+\frac{D}{f}=\frac{D}{\infty}+\frac{D}{f}=\frac{D}{f}$$

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## TAG PROBLEMS

**P1.** **An object is placed at a distance of 15 cm, from a convex mirror and image is formed at a distance of 5 cm from the mirror. Calculate the radius of curvature of the mirror.**

**Sol:** Here, object distance u = –15cm, Image distance v = +5cm

From mirror formula $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}=\frac{1}{5}+\frac{1}{(-15)}=\frac{2}{15}\Rightarrow f=\frac{15}{2}=7.5\,cm;$

$\therefore$ Radius of curvature R = 2f = 2 × 7.5 = 15 cm

---

**P2.** **Calculate the refractive index of material of an equilateral prism for which angle of minimum deviation is 60°.**

**Sol:** Angle of prism $A=60^{\circ}$, Angle of minimum deviation $D_m=60^{\circ}$

Refractive index of prism $\mu=\frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}\Rightarrow\mu=\frac{\sin\left(\frac{60^{\circ}+60^{\circ}}{2}\right)}{\sin\left(\frac{60^{\circ}}{2}\right)}=\frac{\sin 60^{\circ}}{\sin 30^{\circ}}=\frac{\sqrt{3}}{2}\times\frac{2}{1}=\sqrt{3}=1.732$

---

**P 3.** **A convex lens has 20 cm focal length in air. What is its focal length in water? (Refractive index of air-water = 1.33 refractive index for air-glass = 1.5.)**

**Sol:** **Lens maker's formula :** $\frac{1}{f}=(n_{21}-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

(i) When the lens is in air, f = +20 cm, $n_{21}=\frac{n_2}{n_1}=1.5,$

$\therefore\frac{1}{20}=(1.5-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)=(0.5)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$.....(1)

(ii) When the lens is put in water, $n_{21}=\frac{n_2}{n_1}=\frac{1.5}{1.33}$, f = ?

$\therefore\frac{1}{f}=\left(\frac{1.5}{1.33}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$ .....(2)

Dividing equ (1) by equ (2), we get $\frac{f}{20}=\frac{0.5}{\left(\frac{1.5}{1.33}-1\right)}\Rightarrow f=\frac{20\times0.5\times1.33}{1.5-1.33}=\frac{13.3}{0.17}=78.2$ cm

Thus, when a lens is immersed in water, its focal length becomes roughly 4 times to its focal length in air.

☞ Few more **Tag Problems** suggested by your **Faculty.**



# 3 NUCLEI

**8 Marks**

**10. Explain nuclear fission and fusion with suitable nuclear reactions.** **[BMP]**

**A :** **I) Nuclear fission** is the splitting of a heavy nucleus into two or more fragments with the release of large energy .

1) It is the source of energy in nuclear reactors.
2) It takes place at ordinary temperature.
3) This principle is used in atom bomb.
4) The energy produced in the fission is low when compared to fusion.

**Ex:** The fission of Uranium-235 : $_{92}U^{235}+{}_0n^1\rightarrow{}_{92}U^{236}\rightarrow{}_{56}Ba^{144}+{}_{36}Kr^{89}+3\,{}_0n^1+200MeV$

**II) Nuclear fusion** is the fusing of two or more light nuclei to form a single nucleus with the release of extremely large energy.

1) It is the source of energy in the stars.
2) It takes place at very high temperature of the order $10^7$ K.
3) This principle is used in hydrogen bomb.
4) The energy produced in the fusion is 7 times that of fission.

**Ex** : In the Sun 4 hydrogen nuclei fuse together to form a helium nucleus.

$4\,{}_1H^1+4\,{}_{-1}e^0\rightarrow{}_2He^4+2\,{}_{-1}e^0+2\nu+6\gamma+26.7MeV$

---

**11. a) What is the source of stellar energy? Explain it with the help of proton-proton cycle.**
**b) Briefly explain the formation of red giant stage of sun.**

**A:** **a) Source of stellar Energy:** The energy from the sun and other stars is called stellar energy. The source of stellar energy is thermo nuclear fusion.

Nuclear fusion in Sun, stars takes place in multi steps in which the hydrogen is burn to form helium.

**Proton-Proton Cycle:** The nuclear reactions of proton-proton cycle are as follows.

$${}_1H^1+{}_1H^1\rightarrow{}_1H^2+{}_{+1}e^0+\bar{\upsilon}+Q_1$$
$${}_1H^1+{}_1H^1\rightarrow{}_1H^2+{}_{+1}e^0+\bar{\upsilon}+Q_2$$
$${}_1H^2+{}_1H^1\rightarrow{}_2He^3+Q_3$$
$${}_1H^2+{}_1H^1\rightarrow{}_2He^3+Q_4$$
$${}_2He^3+{}_2He^3\rightarrow{}_2He^4+{}_1H^1+{}_1H^1$$

$$4\,{}_1H^1\rightarrow{}_2He^4+2({}_{+1}e^0)+2\bar{\upsilon}+26.7\,MeV$$

Thus 4 hydrogen atoms combine to form a helium atom with a release of 26.7 MeV energy.

**b) Red giant stage of the Sun:**

After about **5 billion years**, most of the **hydrogen fuel** in the Sun's core gets exhausted.

- Then nuclear fusion in the core **slows down**. The **core contracts** and becomes **hotter**.
- Hydrogen starts fusing in a **shell around the core**.
- This releases a lot of energy, causing the **outer layers to expand**.
- As the Sun expands, its **surface cools** and looks **red**.

Thus, the Sun becomes a **Red Giant**—a **very large, cool, and bright (luminous) star**.

12. a) Explain the composition of nucleus and express the atomic masses of nucleons in terms of atomic mass units.

b) Draw and explain the potential energy curve of pair of nucleons.

Show that the density of nucleus doesn't depend on mass number A.

A: a) **Composition of Nucleus:**

- Nucleus consists of positively charged protons and neutrons.
- Charge of proton = $+1.6\times10^{-19}$ C
- The number of protons in the nucleus is called the **atomic number** (Z)
- The no.of protons and neutrons in the nucleus is called **mass number A.**
- **Atomic Masses of Nucleons** in **atomic mass unit (u)**:

**Mass of Proton:** $m_p$=1.007276 u

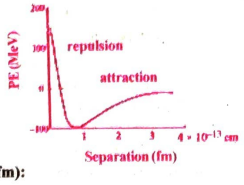
**Mass of Neutron:** $m_n$=1.008665 u

b) **Potential energy curve of nucleons:**

The PE curve shows how potential energy between two nucleons changes with distance between them.

X-axis →Separation between two nucleons in fm.

Y-axis→ Potential energy in MeV.




**The graph shows the following facts :**

i) **Very small distances - Repulsive region (<0.5fm):**

Nuclear force is strongly repulsive. Potential energy is large and positive.

ii) **Intermediate distances - Attractive region (0.5 to 2 fm):**

Nuclear force is attractive. Potential energy is negative.

**Point of stable equilibrium** is attained in this region.

iii) **Large distances (2 to 3 fm):**

Nuclear force becomes very n :gligible. Potential energy becomes almost zero.

- **Derivation of density (ρ):**

Mass of 'A' nucleons M = mA, where m= mass of each nucleon

**Volume of the nucleus is** $V=\frac{4}{3}\pi R^3=\frac{4}{3}\pi R_0^3 A \quad [\because R=R_0A^{1/3}]$

∴ **Density of Nucleus is** $\rho=\frac{M}{V}=\frac{mA}{\frac{4\pi}{3}R_0^3A}=\frac{3m}{4\pi R_0^3}$

Thus, the **density of nucleus** doesn't depend on mass number A.



13. **Define mass defect and binding energy. How does binding energy per nucleon vary with mass number? What is its significance?**

A: 1) **Mass Defect (ΔM)** is the difference between the total mass of all the nucleons of a nucleus and the actual mass of a nucleus (M) .

**Formula:** $\Delta M = [Zm_P + (A-Z)m_n]-M$ where $Zm_P$ = total mass of protons, $(A-Z)m_n$ = total mass of neutrons, M = mass of nucleus.

2) **Binding Energy** is the energy required to bind the nucleons in the nucleus or the energy required to break the nucleus into its constituent protons and neutrons.

From $E = mc^2$ we can write Binding energy $B.E = \Delta M\, c^2$

Here ΔM is the mass defect and c is the velocity of light. Also 1 amu = 931.5 MeV

∴ B.E = ΔM × 931.5 MeV

3) **Binding Energy per Nucleon** ($\overline{B.E}$) is the binding energy of a nucleus (B.E) divided by its mass number (A) . **Formula** $\overline{B.E}$ = B.E/A

4) **Variation of binding energy per nucleon with mass number** :

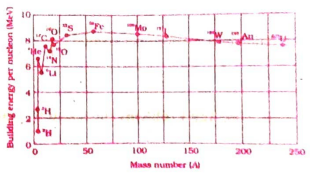
The graph shown in the figure explains how the binding energy per nucleon ($\overline{B.E}$) varies with mass number (A).

**5) The graph shows the following facts :**

i) $\overline{B.E}$ is constant from A = 28 to A = 138

ii) $\overline{B.E}$ =8.75 MeV is maximum for Fe-56 and $\overline{B.E}$ =7.6 MeV lowest for U-238.

iii) $\overline{B.E}$ is low for both light nuclei (A < 28) and heavy nuclei (A > 138).




6) **Significance**: Greater binding energy per nucleon means greater stability to the nucleus.

The light nuclei of A < 28 and heavy nuclei of A > 138 are unstable because their $\overline{B.E}$ is low.

That is why, properties like radioactivity, fission, fusion exist in this region only.

The mid nuclei are stable because their $\overline{B.E}$ is high.

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## TAG PROBLEMS

**P1.** **Find the radius of Aluminum nucleus $_{13}Al^{27}$. (Given that $R_0 = 1.2 \times 10^{-15}$m).**

**A:** Given constant $R_0 = 1.2 \times 10^{-15}$m

From $_{13}Al^{27}$, we have mass number of Aluminium is A=27

Radius of Aluminium is $R = R_0A^{1/3} = 1.2 \times 10^{-15}(27)^{1/3} = 3.6 \times 10^{-15}$ m

---

**P2.** **If one microgram of $_{92}U^{235}$ is completely destroyed in an atom bomb, how much energy will be released?**

**A:** Mass of uranium destroyed, m =1 microgram = $10^{-6}$g=$10^{-9}$kg, c=$3\times10^{8}ms^{-1}$, E = ?

**Formula:** Energy $E = mc^2 = 10^{-9}\times(3\times10^{8})^2 = 9\times10^{-9+8+8}$ $\therefore E = 9\times10^{7}$ J

---

**P3.** **200MeV energy is released when one nucleus of $_{92}U^{235}$ undergoes fission.**

**Find the number of fissions per second required for producing a power of 1 megawatt.** **[BMP]**

**A:** Required power to be produced P = 1 mega watt = $10^6$ J/s

Energy released for per fission in E= 200 meV = $200\times10^{6}\times1.6\times10^{-19} = 3.2\times10^{-11}$J

We know power $P=\frac{NE}{t}\Rightarrow\frac{P}{E}=\frac{N}{t}$

Number of fissions per second $=\frac{P}{E}=\frac{10^6}{3.2\times10^{-11}}=3.125\times10^{16}$

☞ Few more **Tag Problems** suggested by your **Faculty.**




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When Reading becomes Boring, BBQ makes it Caring!

AMAZING MIDDLE SECTION

# SAQ

## SECTION-C

When Exam comes near, BBQ removes Fear!

## SAQ CHAPTERS

**32 MARKS**


| | | | |
|---|---|---|---|
| ★★★ | 1. | ELECTRIC CHARGES AND FIELDS (4 Marks) | 32 - 34 |
| ★★★ | 2. | ELECTROSTATIC POTENTIAL (4+4 Marks) | 35- 39 |
| ★★★ | 3. | CURRENT ELECTRICITY (4 Marks) | 40 - 42 |
| ★★★ | 4. | MAGNETISM AND MATTER (4 Marks) | 43 - 44 |
| ★★½ | 5. | ELECTROMAGNETIC INDUCTION (4 Marks) | 45 - 46 |
| ★★ | 6. | ALTERNATING CURRENT (4 Marks) | 47 - 48 |
| ★★½ | 7. | WAVE OPTICS (4 Marks) | 49 - 50 |
| ★★★ | 8. | DUAL NATURE (4+4 Marks) | 51- 54 |
| ★★★ | 9. | ATOMS (4 Marks) | 55 - 57 |
| ★★½ | 10. | SEMICONDUCTOR ELECTRONICS (4 Marks) | 58 - 60 |


**Super Scoring Pages**

★★★SSP★★★

**Page** 32, 33, 35, 36, 37, 40, 43, 45, 46, 48, 49, 51, 52, 54, 55, 59

**DEAR DUDE!**

Please note that SAQs in each chapter are Jumbled, and Easy SAQs are brought to the beginning.